Lost & Found Platform with Retrieval and Ownership Verification

Document	Software Requirements Specification
Standard	IEEE 830-1998
Project	Single-developer academic research project

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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for a Lost & Found platform built to investigate two research questions in cross-modal retrieval and controlled information disclosure. It defines the behaviour the system must exhibit and the criteria by which each requirement can be tested. It is written for the project supervisor and examiners, and for the developer implementing the system.

1.2 Scope

The system is a web application on which a **finder** registers a found object and a **potential owner** searches for and claims it. Two properties distinguish it from a conventional lost-property noticeboard.

Retrieval is semantic and cross-modal. An owner's free-text description, written from memory, is matched against the finder's photograph in a shared embedding space, then re-ranked using structured metadata. Keyword matching is not used.

Claiming is gated by metered disclosure. Public listings withhold identifying detail. Ownership is assessed by challenge-response, and the information the challenges themselves reveal is priced and capped.

Excluded from scope: payments and rewards, physical custody tracking, hardware tags, mobile applications, multi-institution deployment, and production identity management. Out-of-band notification is also excluded. There is no email or SMS channel; participants see challenges, outcomes and status changes when they next open the site, and every requirement below is written on that basis.

1.3 Definitions, acronyms, and abbreviations

Term	Definition
CLIP	Contrastive Language-Image Pre-training. A model producing image and text embeddings in one shared vector space.
Embedding	A fixed-length numeric vector representing an image or a text passage.
Description gap	Finder and owner describe the same object in non-overlapping vocabulary, so lexical matching fails on a true match.
Proof gap	A public listing reveals enough detail for a fraudulent claimant to construct a convincing claim.
Private attribute	An item detail recorded by the finder at intake, never published, used as challenge ground truth.
Challenge	A question put to a claimant, generated from a private attribute.
Disclosure budget	A numeric cap on how much private information the challenge process may expose, held per claimant and cumulatively per item.
Disclosure freeze	The state after item closure in which no further budget may be spent.
Soft deletion	Marking a record inactive while retaining the row, rather than deleting it.
Idempotent	An operation that, repeated with identical inputs, causes no further state change.
ABO	Amazon Berkeley Objects. A public image dataset used here only as retrieval distractors.
R@k, MRR	Recall at rank k; Mean Reciprocal Rank. Retrieval metrics.
FR / NFR	Functional / Non-Functional Requirement.
[TBD] / [TBC]	To be determined (not yet decided) / To be confirmed (not yet checked against the code).

1.4 References

Ref	Item
R-1	IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.
R-2	Radford et al., *Learning Transferable Visual Models From Natural Language Supervision* (CLIP).
R-3	Amazon Berkeley Objects dataset. Distractor source for the scale ablation.
R-4	Project design record: closure protocol, claim transition table, SDLC selection.
R-5	Companion specification v2.1, IEEE 830-1998 Annex A.5. Full requirement set, threat model, traceability matrix.

1.5 Overview

Clause 2 describes the product informally and states nothing binding. Clause 3 contains every requirement. Appendix A lists unresolved decisions; Appendix B states how the requirements are validated.

Requirements are classified by stability, following the standard distinction:

- **Enduring** requirements derive from the problem domain and are not expected to change. Most of clause 3.2 is enduring: an item must be registered, a claim must be verified, closure must be authorised.
- **Volatile** requirements are expected to change as the project's measurements come in. These are marked **(V)** in the tables. They are mostly thresholds and weights, which will be set by evaluation rather than chosen in advance.

Numbering is inherited from the companion specification. Requirement identifiers are therefore stable but not contiguous, and the gaps carry no meaning. The claim transition table in §3.7 is the one exception: it is reproduced complete rather than selective, for the reason given there.

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2. Overall Description

2.1 Product perspective

The system is self-contained. It has no runtime dependency on any external service; the retrieval model is loaded locally.

Two problems motivate the design, and the design position is that they are separate problems rather than one.

The description gap. An owner writes “silver keyring with a little enamel bird on it”. The finder wrote “metal keys, 3, with blue and orange charm”. The same object, almost no shared words. Keyword search scores this at zero, exactly as it scores two unrelated objects, because the mismatch is conceptual rather than morphological. The system therefore matches the owner’s text against the *photograph*, which takes the finder’s vocabulary out of the matching path altogether. Embeddings then introduce a second problem: they capture category well and instance identity poorly, so five black water bottles sit almost on top of one another in the vector space. Where and when an item was found is not encoded in the photograph at all and comes from a controlled vocabulary, which makes it orthogonal evidence. Re-ranking on it separates cases the embedding cannot.

The proof gap. A listing detailed enough for an owner to recognise is detailed enough for a stranger to recite. Publishing makes items findable and simultaneously destroys the basis for proving ownership; publishing less reverses the trade rather than resolving it. The system resolves it by partitioning item information into four classes rather than reducing it uniformly:

Class	Visible to	Example
Public	Anyone	Category, approximate zone and date, non-identifying photograph
Internal	Retrieval subsystem only	Embeddings, similarity scores
Withheld	Nobody, until verification succeeds	Serial number, contents, distinguishing marks
Controlled-release	One claimant, at a metered cost	What the challenges themselves imply

The fourth class is the one that is easy to miss. Asking “what is written on the back?” reveals that something is written on the back. Verification leaks information, so the system prices each challenge and caps the total.

The retrieval subsystem is advisory throughout. It ranks; it does not decide. No claim or closure is ever authorised by a similarity score.

2.2 Product functions

ID	Function
F-1	Register a found item with photograph, public description, metadata, and private attributes.
F-2	Retrieve ranked candidates from a free-text description.

ID	Function
F-3	Create and progress a claim through a formal state machine.
F-4	Generate challenges, score responses, and decide verification outcomes.
F-5	Meter and cap disclosure, per claimant and cumulatively per item.
F-6	Close an item by handover, withdrawal, or expiry.
F-7	Record every state change in an append-only event history.

2.3 User characteristics

Class	Description	Privileges
Finder	Registers found property. Must record private attributes accurately at intake, which is an unusual demand: the finder never sees them used unless a claim arrives.	Owns registered items; may close them. Cannot reopen.
Claimant	Searches and claims. Usually under time pressure, which lowers tolerance for friction.	Reads public item data; writes only their own claims.
Administrator	Adjudicates disputes, handles budget exhaustion, reopens items closed in error.	All other privileges, plus exclusive reopen and adjudication authority.
Scheduler	Non-human. Sweeps for aged items.	Closure on the expiry path only.

The claimant is at once the actor the security model distrusts and the actor the usability requirements must serve. Every constraint placed on claimants is a cost paid by legitimate owners.

2.4 Constraints

ID	Constraint
C-1	CPU-only inference. No GPU. Embedding costs roughly 100-200 ms per image.
C-2	Single node, single process. No distributed coordination.
C-3	Small corpus: about 60 primary items plus about 2,500 distractors.
C-4	Solo developer, 8-12 week schedule. Production concerns are out of scope by construction.
C-5	File-backed relational store with limited write concurrency.
C-6	Small evaluation sets (about 180 description-image pairs). Statistical power is limited and must be reported as such.
C-7	Distractors are studio catalogue photographs; the primary set is informal phone photography. A model may separate them on image style alone, which would inflate the ablation result. This must be declared wherever the ablation is reported.

2.5 Assumptions and dependencies

ID	Assumption	If false
A-1	The description gap is real and measurable in this domain.	RQ1 loses its motivation. Tested by the token-overlap measurement in Appendix B.
A-2	Pre-trained CLIP transfers to informally photographed everyday objects without fine-tuning.	Baseline retrieval performs near chance.
A-3	Finders record private attributes honestly and accurately.	Challenge ground truth is corrupt and verification scores are meaningless.
A-4	Owners remember at least some private attributes of their own property.	The protocol rejects true owners too often.
A-5	Attribute weights can be assigned defensibly.	The budget becomes an arbitrary counter rather than an information bound.

Assumptions A-2 to A-5 are not verified by any requirement below. Conclusions drawn from the research questions are conditional on them.

Dependencies: pre-trained CLIP ViT-B/32 weights (required at startup); an approximate string-matching library (required for scoring); the photographic dataset and ABO distractors (required for evaluation only, not for operation).

3. Specific Requirements

3.1 External interface requirements

3.1.1 User interfaces

Server-rendered pages, accessed through a standard browser. No installation.

ID	Requirement
UI-001	The registration form shall present the private-attribute section as visually and textually distinct from public fields, and shall state why those attributes are collected.
UI-004	Candidate lists shall show public-class fields only, with ranking position.
UI-010	The challenge interface shall not display remaining challenges, remaining budget, or any progress indicator derived from budget state.
UI-013	The closure action shall require a reason code and an explicit confirmation before executing.
UI-015	Handover shall present separate confirmations to finder and claimant, each showing which confirmations are outstanding.
UI-016	The item timeline shall render chronologically, scoped to the viewer's permitted visibility.

3.1.2 Hardware interfaces

None. Photographs enter as uploaded files.

3.1.3 Software interfaces

Component	Purpose
CLIP ViT-B/32, pre-trained	Image and text embedding
Approximate string matcher	Challenge response scoring
File-backed relational store	Transactional persistence

The retrieval subsystem takes a text query plus metadata and returns an ordered list of `(item_id, baseline_score, final_score)` restricted to active items. It is side-effect free with respect to item and claim state, deterministic, and holds no authority over any state transition.

3.1.4 Communications interfaces

HTTP over a local network. The system makes no outbound calls at runtime. The one endpoint fixed by the project is `POST /items/{item_id}/close`, which requires an authenticated finder or administrator and a reason code, and responds with a redirect to the item view. The enumeration of valid reason codes is [TBD]; `withdrawn` and `expired` are settled, and the handover path implies a third.

3.2 Functional requirements

Identifiers are inherited from the companion specification. **(V)** marks a volatile requirement whose threshold or weighting is expected to change once measurements are available.

3.2.1 Item management

ID	Requirement	Verification
FR-010	The system shall allow an authenticated finder to register a found item.	Register; assert persisted.
FR-011	The system shall require at least one photograph and the metadata fields category, zone, and date found.	Submit incomplete; expect rejection.
FR-013	The system shall assign every item information field to exactly one of the four disclosure classes.	Inspect field mapping for total, exclusive coverage.
FR-014	The system shall store private attributes in the withheld class and shall never render them in a public view.	Fetch as an unrelated user; assert absence.
FR-015	The system shall assign a newly registered item the status active.	Register; assert status.

ID	Requirement	Verification
FR-016	The system shall generate and persist an embedding for each item image.	Register; assert embedding non-null.
FR-021	The system shall reject any update to an item whose status is not active.	Update a resolved item; expect rejection.
FR-024	The system shall prevent modification of a private attribute once a challenge has been issued from it.	Issue challenge; attempt edit; expect rejection.
FR-028	The system shall exclude non-active items from all retrieval candidate sets.	Resolve an item; re-run a query that previously returned it; assert absent.

3.2.2 Matching and retrieval

ID	Requirement	Verification
FR-030	The system shall accept a free-text description as a retrieval query.	Submit query; assert accepted.
FR-031	The system shall embed the query into the same vector space as item images.	Assert shared dimensionality.
FR-032	The system shall compute a baseline cosine similarity against every active item.	Assert a score per active item.
FR-039 **(V)**	The system shall signal low confidence where no candidate exceeds the viability threshold.	Query with unrelated text; assert indication. [TBD] threshold.
FR-041	The system shall apply a metadata re-ranking layer over baseline similarity.	Assert final score differs from baseline where metadata differs.
FR-042 **(V)**	The system shall compute metadata agreement over category, location zone, and date proximity.	Assert each dimension contributes. [TBD] weights.
FR-044	The system shall produce identical rankings for identical inputs.	Repeat query; assert identical order.
FR-045	The system shall support execution with re-ranking disabled, yielding the semantic-only baseline.	Run both modes; assert both produce output.
FR-046	The system shall record the ground-truth rank under both configurations for each evaluation query.	Assert both ranks in evaluation output.
FR-047	The system shall grant no claim right, verification credit, or authorisation on the basis of a retrieval score.	Assert claim and verification paths take no similarity input.

FR-045 and FR-046 serve the research question rather than the user. Without a selectable baseline mode and paired rank recording, no ablation is possible. FR-047 keeps retrieval advisory: if a high match score earned verification credit, an attacker could gain it by writing a description that matches well, which is precisely what the public listing enables.

3.2.3 Claim lifecycle

ID	Requirement	Verification
FR-050	The system shall allow a user to claim an active item they did not register.	Create claim; assert persisted.
FR-052	The system shall permit at most one open claim per claimant per item.	Create a second; expect rejection.
FR-054	The system shall permit only transitions declared legal in the transition table (§3.7).	Attempt each legal transition; assert success.
FR-055	The system shall reject any undeclared transition with an explicit illegal-transition error.	Attempt each illegal transition; assert error.
FR-056	The system shall leave state unchanged when a transition is rejected.	Assert state identical to before.
FR-057	The system shall enforce the authorised actor for each transition.	Attempt each transition as each unauthorised role.
FR-058	The system shall serialise concurrent state changes on one item using row-level locking.	Concurrent transitions; assert exactly one succeeds.
FR-062	The system shall close all other open claims when an item is resolved, in the same transaction as the status change.	Resolve with several open claims; assert all closed atomically.
FR-064	The system shall route indeterminate verification outcomes to	Force indeterminate

ID	Requirement	Verification
	administrator adjudication rather than defaulting to approval or rejection.	outcome; assert routed.
FR-065	The system shall require a justification when an administrator records a final disposition on a claim.	Adjudicate without one; expect rejection.

3.2.4 Verification and disclosure control

ID	Requirement	Verification
FR-070	The system shall generate challenges from the withheld attributes of the claimed item.	Assert challenges derive from recorded attributes.
FR-071	The system shall not generate a challenge from any public-class information.	Inspect challenges; assert no public field is the subject.
FR-072 **(V)**	The system shall assign each private attribute a weight representing its discriminative value.	Assert every attribute carries a weight. [TBD] scheme.
FR-073	The system shall derive each challenge's disclosure cost from the weight of the attribute it references.	Assert cost recorded per challenge.
FR-074	The system shall select challenges such that the issued set's total cost does not exceed the claimant's remaining budget.	Attempt an over-budget issue; assert refused.
FR-077 **(V)**	The system shall score responses by approximate string matching, tolerating minor variation in spelling, case, and phrasing.	Submit near-matches; assert scored as matching. [TBD] threshold.
FR-078	The system shall compute the verification score as matched weight over issued weight.	Assert score equals the documented formula.
FR-079 **(V)**	The system shall decide pass or fail by comparing the score against a configured threshold.	Assert outcome derives from the comparison. [TBD] threshold, to be calibrated by ROC analysis.
FR-081	The system shall treat an empty or unparseable response as non-matching, without halting the scoring of the remaining responses.	Submit malformed input; assert graceful handling.
FR-082	The system shall debit the per-claimant budget by each challenge's cost at issue, whether or not the claimant responds.	Issue without responding; assert debited.
FR-083	The system shall debit the item's cumulative budget by the same cost in the same transaction.	Assert both counters move together.
FR-085	The system shall halt automated claiming and route the item to manual review when its cumulative budget is exhausted.	Exhaust across claimants; assert halted.
FR-086	The system shall not replenish either budget on rejection, expiry, or withdrawal.	Reject a claim; assert counters unchanged.
FR-087	The system shall not reset the cumulative budget when a new claimant arrives.	File a second claim; assert carry-over.
FR-089	The system shall never expose budget state to a claimant.	Fetch claim view; assert no budget field.

Attribute weight does double duty: it measures how much an attribute proves and how much disclosing it costs. The attributes most useful for establishing ownership are exactly the ones most damaging to reveal, so one number governs both, and that coupling is the mechanism.

FR-082 charges at issue rather than at response, because the information leaves the system when the question is asked. Charging on response would let an attacker request challenges, read what they imply, abandon the claim, and repeat at no cost. FR-086 and FR-087 make this a bound rather than a rate limit: a budget that refunds or resets is not a budget, and the sequential-claimant attack, where someone files claims under successive identities until one accumulates enough to pass, is defeated only by monotonic cumulative consumption.

The mechanism does not prove ownership. It raises the cost of a fraudulent claim from reading a listing to holding knowledge that was never published. It cannot distinguish an

owner from a thief, or from anyone else with genuine prior knowledge. Its output is a score and a recommendation; contested cases go to a human.

3.2.5 Closure and event history

ID	Requirement	Verification
FR-090	The system shall support exactly three closure paths: verified two-party handover, finder withdrawal, and scheduled expiry.	Assert each reachable; assert no fourth.
FR-091	The system shall permit closure only by the finder, an administrator, or the scheduler on the expiry path.	Attempt as claimant; expect rejection.
FR-093	The system shall require a reason code on every closure.	Close without one; expect rejection.
FR-094	The system shall set status to resolved and record a resolution timestamp on closure.	Close; assert both.
FR-097	The system shall freeze the item's disclosure budget on closure.	Close; attempt challenge issue; expect refusal.
FR-098	The system shall make closure idempotent: a repeated request shall not alter state, append a second event, or produce an inconsistent state.	Close twice; assert one state change and one event.
FR-101	The system shall perform the status change, cascade closure, and budget freeze in one atomic transaction.	Inject mid-operation failure; assert full rollback.
FR-104	The system shall require confirmation from both finder and verified claimant before completing handover.	Confirm as one party; assert not closed.
FR-112	The system shall not expire an item with a claim in verification.	Age such an item; run sweep; assert not closed.
FR-113	The system shall permit reopening only by an administrator, with a justification.	Attempt as finder; expect rejection.
FR-116	The system shall restore a reopened item's budget to its frozen value, not to a fresh allowance.	Consume, close, reopen; assert consumed total preserved.
FR-120	The system shall record every state-changing operation in an append-only event history.	Enumerate operations; assert each produces an event.
FR-123	The system shall record event type, affected entity, acting identity, and timestamp on every event.	Assert all four present and non-null.
FR-133	The system shall write each event in the same transaction as the state change it records.	Inject post-change failure; assert both rolled back.

Single-party closure is the main fraud route: an impostor claims an item, marks it received, the item leaves the pool, and the true owner never sees it. FR-104 makes the party who would lose property a required participant in losing it. FR-116 blocks a subtler attack. If reopening restored a fresh allowance, an administrator persuaded to reopen an item would silently undo an exhausted budget.

3.3 Performance requirements

ID	Requirement	Target
NFR-PERF-001 **(V)**	Retrieval shall return a ranked candidate set over the full active corpus within an interactive response time.	[TBD], to be set after measurement at target scale.
NFR-PERF-002	Retrieval shall run without GPU acceleration.	CPU-only.
NFR-PERF-003	Image embedding shall complete within a time acceptable at registration, or be deferred without blocking it.	About 100-200 ms per image.
NFR-PERF-005	Retrieval time shall scale linearly with active corpus size, and the scaling shall be measured rather than assumed.	Linear, with the crossover point identified.
NFR-PERF-006	Resolved items shall be filtered out before similarity scoring, so they impose no scoring cost.	Zero cost.
NFR-PERF-S2	The system shall support an active corpus of at least 2,600 items.	60 primary plus about 2,500 distractors.

NFR-PERF-005 is where the decision not to use an approximate index is defended. Brute-force search is linear per query; at roughly 2,500 items with 512-dimensional vectors that is one small matrix multiplication, comfortably interactive. An index would add a build step, a rebuild obligation, a tuning surface and a dependency, for no gain at this scale. The requirement is to *demonstrate* that and to identify the corpus size at which the trade-off inverts, which is a stronger claim than asserting the index is unnecessary.

3.4 Logical database requirements

Nine logical entities, listed below.

Entity	Purpose	Notable fields
User	Authenticated identity. Roles are contextual: finder for items registered, claimant for claims filed.	user_id, role
Item	A registered found object.	status, resolved_at, embedding, closure_reason
PrivateAttribute	Withheld detail serving as challenge ground truth. The most sensitive entity in the system.	ground_truth_value, weight
Claim	An ownership assertion and the container for verification.	state, closure_cause
Challenge	A question generated from a private attribute.	question_text, disclosure_cost
ChallengeResponse	A claimant's answer and its match outcome.	response_text, match_outcome
VerificationRecord	The persistent outcome of verification for a claim.	score, threshold_applied, outcome
DisclosureBudget	The information bound. Exists at two scopes: per claim and per item.	allowance, consumed_total, frozen
ItemEvent	Append-only audit record. Never updated, never deleted.	event_type, actor_id, timestamp

Field semantics. `status` is simultaneously a lifecycle marker, a retrieval-eligibility predicate, a mutation guard and a disclosure gate, which makes it the highest-consequence field in the schema. `resolved_at` marks the boundary after which no disclosure may be attributed to the item. The closure reason code is what makes the event history interpretable: an item that is merely "closed" says nothing about whether property was returned, withdrawn, or timed out, and those three cases mean entirely different things.

Retention. Closure is soft deletion. Rows are retained because hard deletion would break referential integrity with claims and events, destroy the audit trail, and discard resolved cases, which are the only human-verified positive match pairs the system produces.

Integrity invariants.

ID	Invariant
INV-1	No claim in a non-terminal state may exist against a resolved item.
INV-2	An item's cumulative consumed disclosure equals the sum of the costs of all challenges issued against it.
INV-5	No item has status resolved with a null resolved_at or a null reason code.
INV-6	Consumed disclosure totals are monotonically non-decreasing over an item's lifetime, including across closure and reopening.

3.5 Design constraints

Requirements say what the system must do. The design decisions below say how this prototype does it, and may change without changing any requirement.

Requirement	Design decision	Constraint

Requirement	Design decision	Constraint
Rank by semantic similarity and metadata (FR-041)	CLIP ViT-B/32; cosine similarity; re-ranking on category, zone, date	CPU-only (C-1)
Interactive retrieval (NFR-PERF-001)	Brute-force cosine over an in-memory matrix; no approximate index	Corpus size makes linear scan tractable (C-3)
Durable transactional storage	SQLite, single file	Single node (C-2). SQLite is *not* a requirement; any transactional store satisfying FR-101 and FR-133 would serve.
Serialise concurrent state changes (FR-058)	Row-level locking within one transaction	Single process (C-2)
Idempotent closure (FR-098)	Guarded update `UPDATE ... WHERE id = ? AND status = 'active'`, branching on affected row count	Requires atomic conditional update
Exclude resolved items (FR-028)	`WHERE status = 'active'` applied before scoring	Filter-before-score so resolved items cost nothing
Declared transition table (FR-054)	One transition dictionary keyed by (state, trigger), raising on absent keys	Chosen so the table is both implementation and test parameterisation
Bounded disclosure (FR-082 to FR-087)	Two monotonic counters debited in one transaction at issue	[TBD] allowance values

3.6 Software system quality attributes

Attribute	ID	Requirement
Reliability	NFR-REL-002	Concurrent state changes on one item shall be serialised.
	NFR-REL-003	State change, cascade, budget accounting, and event write shall be atomic; no partial application shall be observable.
	NFR-REL-004	No claim shall be left in a state inconsistent with its parent item's status.
Security	NFR-SEC-002	Authorisation shall be enforced at the request-handling layer, not by interface omission.
	NFR-SEC-003	Withheld attributes shall never reach a claimant in any view, response body, or error message.
	NFR-SEC-004	No claimant shall observe another claimant's existence, claim, responses, or outcome.
	NFR-SEC-005	The disclosure budget shall be a hard limit; over-budget requests shall be refused, not logged and permitted.
	NFR-SEC-009	Repeated identical state-changing requests shall be safe.
Privacy	NFR-PRIV-001	No item information shall be published beyond what an owner needs to recognise their property.
	NFR-PRIV-004	Total information exposure per item shall be bounded across all claimants for the item's lifetime.
	NFR-PRIV-005	No route shall replenish a budget, including rejection, expiry, new claimant, closure, or reopening.
Auditability	NFR-AUD-002	No application route shall alter or remove a recorded event.
	NFR-AUD-004	An item's full history, including every disclosure, shall be reconstructable from the event record alone.
Maintainability	NFR-MNT-001	The retrieval subsystem shall be replaceable without modifying claim, closure, or verification logic.
	NFR-MNT-003	Baseline scoring and metadata re-ranking shall be independently modifiable and disableable.
	NFR-MNT-006	The system shall be testable without network access or GPU hardware.
Usability	NFR-USE-001	Claim state shall be shown to the claimant in plain language with the next required action, not as an internal identifier.
	NFR-USE-002	The registration interface shall explain why private attributes are collected and that their accuracy determines whether the owner can later be verified.
	NFR-USE-005	A rejected claimant shall be told the outcome, without the correct answers or the score breakdown.

NFR-USE-005 costs the system real usability. Good design would tell a rejected claimant what they got wrong; that is exactly the disclosure the system must not make, so the privacy requirement wins. Three gaps are declared rather than concealed: authentication strength, credential storage and session management are [TBD] and outside the research scope; timing side channels are not addressed; and a finder who records unanswerable attributes to retain property is not defended against, since the system assumes finder good faith (A-3).

3.7 Object-oriented models

3.7.1 Data flow diagram (Level 1)

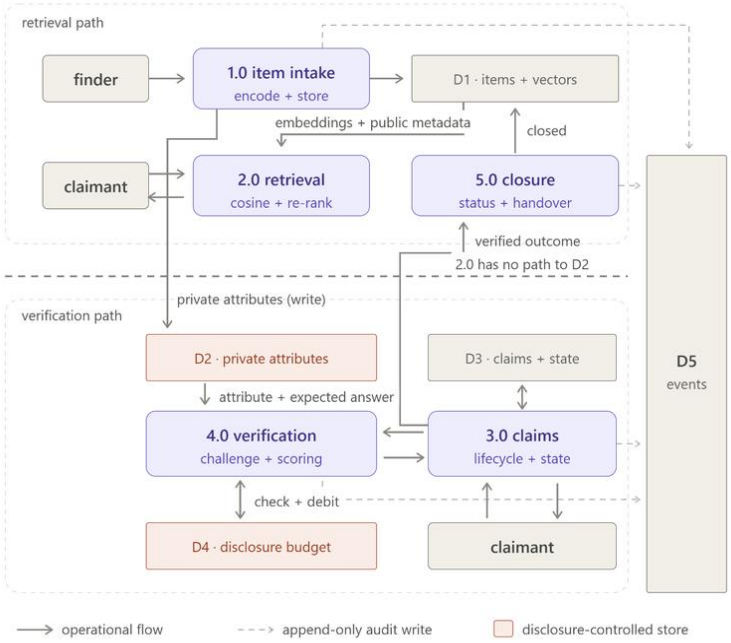


Figure 1 - Level-1 data flow diagram. Solid arrows are operational flows; dashed arrows are append-only audit writes.

D1, items and embeddings, feeds retrieval. D2, private attributes, feeds verification. Nothing flows from D2 into process 2.0, and that absence is the proof-gap treatment drawn rather than argued: what makes an item findable and what proves ownership come from disjoint stores. Process 4.0 is the only reader of D2, and every read is metered against D4.

#	From	Trigger / actor	To	Side effects
T1	-	claim_filed / C	PENDING	Creation event
T2	PENDING	challenge_sent / V	VERIFYING	Both budgets debited
T3	VERIFYING	score_pass / V	VERIFIED	Verification record persisted
T4	VERIFYING	score_fail / V	REJECTED	No budget refund
T5	VERIFYING	timeout / S	EXPIRED	No budget refund
T6	VERIFIED	handoff_done / F + C	CLOSED	Item closed; budget frozen; other claims cascade-closed

#	From	Trigger / actor	To	Side effects
T7	VERIFYING	budget_exhausted / V	MANUAL_REVIEW	Automated claiming halted
T8	MANUAL_REVIEW	admin_approve / A	VERIFIED	Justification required (FR-065)
T9	MANUAL_REVIEW	admin_reject / A	REJECTED	Justification required (FR-065)
T10	PENDING, VERIFYING, VERIFIED	item_resolved_elsewhere / S	CLOSED_ITEM_RESOLVED	Runs inside the closure transaction
T11	PENDING, VERIFYING	claimant_withdraws / C	REJECTED **[TBC]**	No budget refund
T12	VERIFYING	contested / V, A	DISPUTED	Administrator queue entry
T13	DISPUTED	admin_resolve / A	VERIFIED or REJECTED	Justification required (FR-065)

Actors: C claimant, F finder, A administrator, S system or scheduler, V verification subsystem.

Any (state, trigger) pair absent from this table is forbidden and raises an illegal-transition error, leaving state unchanged. The forbidden set is derived by complement, so the two cannot drift apart. That is why this table is reproduced in full while the rest of the document is selective: an omitted row here would be a prohibition, not an abbreviation. Terminal states cannot be escaped, so a rejected claimant cannot retry and consume budget again. No path runs from PENDING to VERIFIED or CLOSED, so verification cannot be bypassed. MANUAL_REVIEW and DISPUTED are the two non-terminal states an automated actor cannot leave; only T8, T9 and T13 exit them, each requiring an administrator and a justification. T10 is the only transition permitted on a claim whose item has been resolved.

[TBC] - Confirmation required. Two things here need checking against the code before submission. The state names come from the project design record and have not been re-verified (OI-01). T11's destination is unsettled: withdrawal currently lands in REJECTED, though a distinct WITHDRAWN state would read better in the evaluation, since a withdrawal is not a failed verification (OI-03). The requirements in §3.2.3 are written to be independent of the identifiers. They constrain that transitions are governed by a declared table, that undeclared ones are rejected, and that actors are authorised. Renaming a state invalidates none of them.

Appendix A - Open items

Unresolved decisions bearing on the requirements in this document. None may be treated as settled. Identifiers are those of the companion specification, which carries the full list, OI-01 to OI-32.

ID	Item	Clause
OI-01	Claim state identifiers unverified against the codebase	3.7
OI-03	Whether claimant withdrawal (T11) reaches REJECTED or a distinct WITHDRAWN state	3.7

ID	Item	Clause
OI-07	Metadata combination function and weights	FR-042
OI-08	Metadata applied as a hard filter or a soft prior. A hard filter is more decisive but fails badly when the owner misremembers where they lost the item, which is common. Current position favours the soft prior.	FR-041
OI-09	Attribute weight assignment scheme	FR-072
OI-10	String-similarity threshold for response matching	FR-077
OI-11	Verification decision threshold	FR-079
OI-12	Per-claimant and cumulative budget allowances	FR-082
OI-13	Challenge selection policy under a constrained budget. High-weight attributes first maximise discrimination per unit of budget but leak the most valuable information earliest.	FR-074
OI-14	Expiry threshold and sweep frequency	FR-112
OI-18	Behaviour when embedding fails at registration. Silent acceptance would produce an invisible item and must be excluded.	FR-016
OI-19	Authentication mechanism and session management	3.6
OI-28	Embedding performed synchronously at registration or deferred	NFR-PERF-003

OI-09 and OI-12 are coupled. Attribute weight is both discriminative value and disclosure cost, so weights and budget allowances cannot be decided independently; resolving one without the other produces an incoherent operating point.

Two further [TBD]s in this document carry companion identifiers that have not been reconciled here: the interactive latency target for NFR-PERF-001, and the enumeration of closure reason codes for §3.1.4 and FR-093.

Appendix B - Verification and validation

Unit and state-machine testing. Transition tests are parameterised over the table in §3.7, so adding a transition extends coverage automatically. Three families: every legal transition succeeds; every undeclared (state, trigger) pair raises an error and leaves state unchanged; every transition attempted by an unauthorised role is rejected. Closure tests cover idempotency, authorisation by role, cascade, and atomicity under injected fault.

Description-gap measurement (A-1). Token overlap between each finder's public description and the three independently written owner descriptions of the same object, across all 180 pairs. This is what establishes that lexical matching has little to work with in this domain, and it must be run before the retrieval results are interpreted. If overlap turns out to be high, RQ1 is measuring something other than what it claims.

Retrieval evaluation (RQ1). About 60 photographed objects with three independently written descriptions each, giving about 180 pairs, over three conditions: a lexical baseline, semantic-only, and semantic plus re-ranking. Metrics R@1, R@5, R@10, MRR. Per-query rank changes must be reported **including cases where re-ranking makes the rank worse**, since reporting only the aggregate would hide the layer's failure modes. Performance on visually near-identical clusters is the sharpest test of the claim.

Verification evaluation (RQ2). Simulated legitimate owners, holding the true attributes, against simulated opportunists holding only the public listing. Outputs: score distributions, an ROC curve, a selected threshold with justification, and the **false rejection rate for legitimate owners** at that threshold, which is the cost side and must be reported as prominently as the discrimination. A sequential-claimant simulation asserts that total disclosure never exceeds the item allowance (INV-2).

Scale ablation. Corpus grown from about 60 to about 2,500 items with ABO distractors, measuring retrieval degradation and latency scaling, and identifying the corpus size at which an approximate index would pay for itself (NFR-PERF-005). This is an evaluation activity, not a functional requirement. It measures degradation under corpus *size*, not corpus *difficulty*: the distractors are studio photographs and the primary set is phone photography, so a model may reject them on image style alone and inflate the result (C-7).

No results appear anywhere in this document. None of these evaluations has been run.

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Assumptions, §2.5. Budget freeze, FR-097, FR-116. Cascade closure, FR-062. Challenge generation, FR-070 to FR-074. Claim states, §3.7. CLIP, §1.3, §3.1.3. Closure paths, FR-090. Constraints, §2.4. Data flow diagram, §3.7.1. Description gap, §2.1, Appendix B. Disclosure budget, §3.2.4. Disclosure classes, §2.1, FR-013. Event history, FR-120 to FR-133. Guarded update, §3.5. Idempotent closure, FR-098. Invariants, §3.4. Metadata re-ranking, FR-041, FR-042. Open items, Appendix A. Private attributes, FR-014, FR-024. Proof gap, §2.1. Reopening, FR-113, FR-116. Retrieval, §3.2.2. Row-level locking, FR-058. Scale ablation, NFR-PERF-005, Appendix B. Soft deletion, §3.4. State machine, §3.7. Threshold calibration, FR-079. Two-party handover, FR-104. Volatile requirements, §1.5.